

Glacier's Lament

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ACM Reference Format:

Jiabao Li and Cooper Galvin. 2022. Glacier's Lament. In *Proceedings of the ACM SIGGRAPH '22 Art Gallery*, August 08-11, 2022, Vancouver, BC, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532837.3534948>

1 PROJECT URL

www.jiabaoli.org/glacier

2 DESCRIPTION OF ARTWORK

Glaciers are sentinels of climate change. They are the most visible evidence of global warming today. This series of works embodies the stunning beauty, rapid change, fragility, destructive power, and magnificence of glaciers. At the same time, they challenge the audience with the dramatic, irreversible ecological damages from climate change.

In *Glacier's Lament*, we used data from glacier melting in the past 60 years to compose music and dance with local musicians who have witnessed the recession of the Mendenhall glacier over their lifetimes in Juneau, Alaska. Each note is one season in a year. In the winter, the glacier is frozen, so the pitch is low. In the summer, the melting rate rises, so the pitch is high. Towards the end, the melting overflows into spring and autumn, and the melting in the summer becomes faster. We filmed the artists performing the piece on the glacier, collaborating with the glacier's own sounds.

There are four color cards in PANTONE for glacier blue. However, in real glaciers, this blue color is variable and dynamic. As glaciers are disappearing, this unique blue is also disappearing. We sampled and blended the blue color from glaciers in Alaska and hung them in recycled glass vials. When one glacier calving happened, one color vial fell down. At the end of the exhibition, all 60 vials fell down, forming a painting on the canvas beneath.

3 HEARING GLACIER'S LAMENT

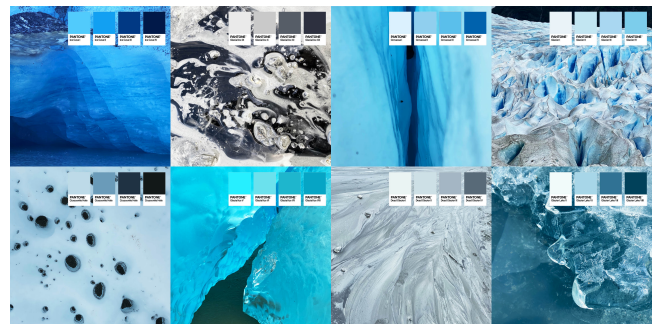
In *Glacier's Lament*, we sought to represent the global scale of ice loss while touching upon the local feeling of loss of the artists who have lived near the glacier their entire lives. We took data of the polar ice sheet melting from 1959 to 2019 and wrote a program in Matlab to convert the data into sound waves, creating a duet. Every measure of the music is a year and every note is a season's data on cumulative ice loss. Summer months are thus high tones, with

winter being the lowest tone in each measure. Two musicians from Juneau converted the sonified data from the software into sheet music and improvised a violin melody atop the computer-generated music, thus combining the personal experience of loss of a familiar face and beautiful natural phenomenon with the data of ice loss.



4 BRINGING THE DISAPPEARING BLUE TO COMMUNITIES FAR OUTSIDE ALASKA

There are four color cards in PANTONE for glacier blue: Glacier Lake (16-4118 TCX), Blue Ice (PQ- 17-3922TCX), Glacier (12-5505 TPX), and Glacier Gray (14-4102 TCX). However, in real glaciers, this blue color is variable and dynamic. As glaciers are disappearing, this unique blue is also disappearing.



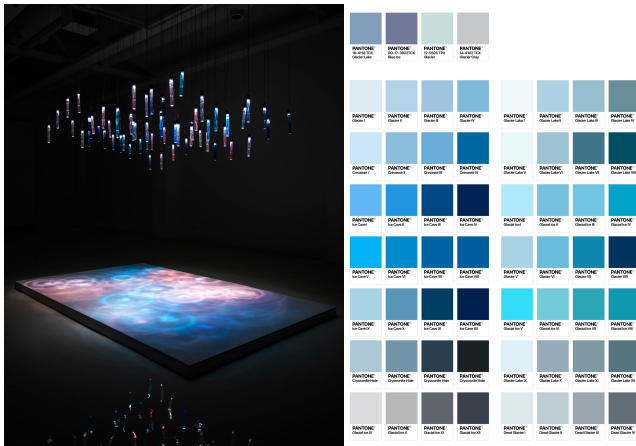
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SIGGRAPH '22 Art Gallery, August 08-11, 2022, Vancouver, BC, Canada

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ACM ISBN 978-1-4503-9372-0/22/08.

<https://doi.org/10.1145/3532837.3534948>



5 CONCLUSIONS

Glacier's Lament attempts to express and meld the feeling of loss of ephemeral beauty of glacier with the factual evidence of climate change. The goal is that in synthesizing these two realities (the factual and the felt), the work can reach audiences beyond those directly impacted by the loss of glaciers before it is too late. At the same time, it is a love letter to an already diminished nature and mourning of that loss. *Glacier's Lament* and *The Disappearing Blue* seek to make the impersonal data personal, and the personal experience of a changing environment heard. They recreate aesthetic dynamics of the loss of nature.

6 ARTIST INFORMATION

Jiabao Li creates works addressing climate change, interspecies world sharing, humane technology, and a just, sustainable future. She is a current member of NEW INC Creative Science at the New Museum. Her work has been exhibited internationally, at Ars Electronica, SIGGRAPH, Milan and Dubai Design Week, Shanghai Ming Contemporary Museum, ISEA, Anchorage Museum, OCAT Contemporary Art Terminal, CHI, Donghu Shan Art Museum, MOOD Museum of Design, Alaska State Museum. Jiabao is an Assistant Professor at UT Austin. Her lab explores the intersection of art, design, technology, and biology. The interests are broad spanning from interspecies co-creation to menstrual blood science. Jiabao graduated from Harvard Graduate School of Design with a Master of Design in Technology with Distinction.

Cooper Galvin is from Anchorage and Juneau, Alaska, where he learned to raise chickens, mush sled dogs, and forage. Cooper studied Chemistry and Mathematics at Pomona College. He then studied Biophysics at Stanford where he also started the science, technology, and art nonprofit: future advancers of science and technology. Since defending and receiving his PhD in January 2020, Cooper did a brief visiting postdoctoral fellowship at Taipei Medical University, consulted for McKinsey and Company, taught computational genomics as an instructor for underrepresented minority students at the Broad Institute of Harvard and MIT, and is now CEO of Peregrine Biosense, a startup specialized in instant virus detection.

ACKNOWLEDGMENTS

We want to thank all the Alaskan artists who contributed to this project: composer and violinist Lindsay Clark, choreographer and dancer Anouk Otsea, cellist James Cheng, and videography director Ryan Cortes Perez. Special thanks to glaciologist Marco Tedesco from Columbia University, who provided the data set and gave us scientific advice. We thank Anchorage Museum, Bloomberg Philanthropies, and Donghu Shan Art Museum, who funded part of the project. We thank our glacier guide Elias Antaya and oceanologist Cathy Botelho and John Clark for their support.